1. Clinical Question & Core Hypothesis

Clinical Question

What molecular mechanisms determine whether immune structures translate into clinical benefit?

Working Hypothesis

Primary hypothesis: Sequential chemotherapy induces upregulation and/or relocalization of TAS2R38-associated signaling in ERBB2-positive breast cancer cells, resulting in increased HER2 cell-surface density and enhanced trastuzumab-mediated ADCC, an effect that is contingent on a high Tumor-Infiltrating Lymphocyte (TIL) density. Secondary predictions: 1) Chemotherapy increases TAS2R38 mRNA/protein levels and downstream signaling in ERBB2+ cells; 2) Post-chemotherapy, HER2 surface density increases in TAS2R38-active cells; 3) In co-culture or organoid systems with high TIL density, chemo-treated, TAS2R38-competent tumor cells exhibit greater trastuzumab-mediated cytotoxicity compared with TAS2R38-deficient counterparts; 4) TAS2R38 knockdown or pharmacologic inhibition diminishes the chemotherapy-associated gain in trastuzumab response; 5) In vivo, tumors that are ERBB2+, TAS2R38-competent, and embedded in a high-TIL microenvironment show superior response to sequential chemotherapy followed by trastuzumab compared with tumors lacking chemo-induced TAS2R38 activation or with low TIL density.

Element	Value
Imaging feature	Tumor-Infiltrating Lymphocyte (TIL) density
Anchor gene(s)	ERBB2, TAS2R38
Primary endpoint	therapy_response

2. Why This Gene — Not Others

Primary hypothesis gene(s)

ERBB2: ERBB2 and trastuzumab sequence; I focused on the mechanistic chain that chemotherapy may induce receptor expression changes that affect HER2 availability.

TAS2R38: TAS2R38 localization → Breast cancer → HER2/ERBB2 and trastuzumab sequence; I focused on the mechanistic chain that chemotherapy may induce receptor expression changes that affect HER2 availability.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	NCBI:9687	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: allows exploration of a potential regulatory node that may influence TAS2R38 signaling or membrane trafficking.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Novel mechanistic hypothesis derived from knowledge graph analysis.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- NCBI:9687: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.